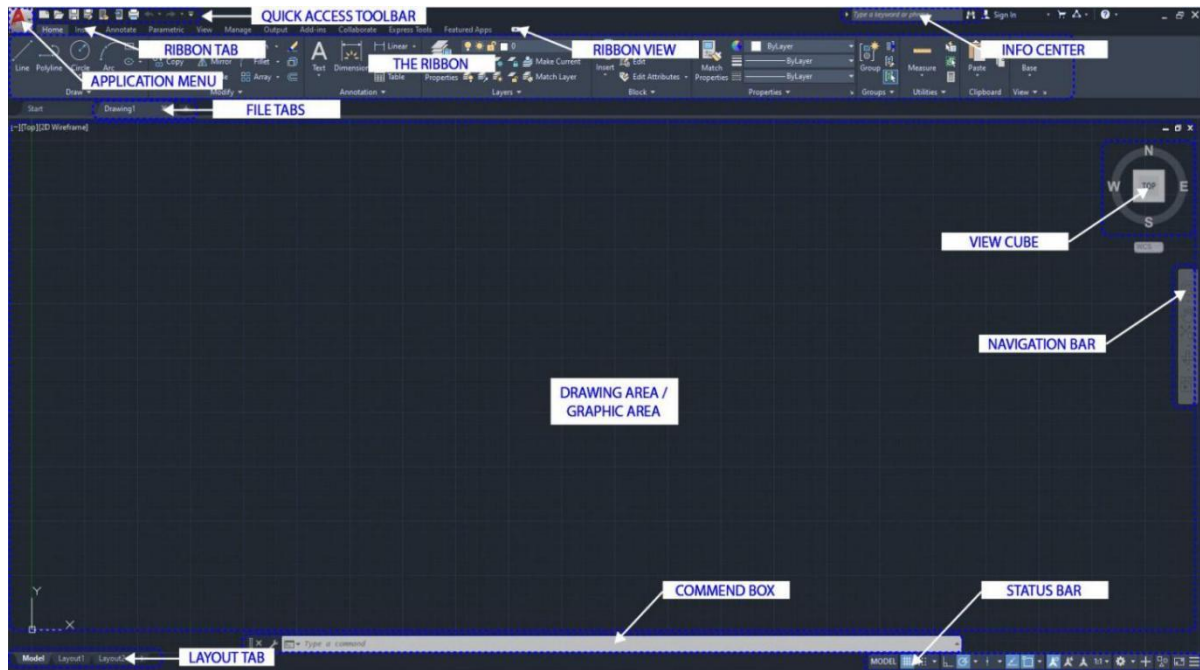


AutoCAD

This is the most commonly-used CAD (computer-aided design) and drafting software. It can be defined as a standard CAD software used to assist in the creation, modification, and optimization of a design. With AutoCAD we can create both 2D and 3D drawings used in construction and manufacturing. It was developed by **John Walker** in **1982** with the help of **AUTODESK** and maintained it successfully. The program is utilized by architects, engineers, graphic designers, city planners, and project managers due to its powerful 3D modelling capabilities. It stands on demand to students and industries because of its specifications & requirements.



AutoCAD Application Interface

- *Application menu*: New, open, save, import, export, print
- *Quick access toolbar*: User can save tools that they often use
- *Info Centre*: Ask a question, find out answers from Autodesk community
- *Ribbon*: Main menus – Home, Insert, Annotate, View, Manage
- *Ribbon tab*: This consist composed of series of tabs organised in panels containing tools to modify drawing.
- *Ribbon view*: User can minimize and maximize the ribbon
- *File tab*: Navigate files and create and open files
- *Drawing area/graphic area*: Main drawing space
- *View cube*: User can change the view, top, front, 3D, or more
- *Navigation bar*: Zoom in and out, pan, zoon to all, and more
- *Command box*: Can type commands and see the previous commands
- *Layout tab*: Can see model space and print spaces
- *Status bar*: Can set grid, snaps, scales, and more

HOW TO INSTALL AUTOCAD SOFTWARE

1. Insert the AutoCAD DVD or CD into your computer's drive.
2. Locate the Drive in your File Explorer
3. In the AutoCAD Installation Wizard, click Install Products.
4. On the Welcome to the AutoCAD--- Installation Wizard page, click Next.
5. Select the product(s) you want to install and click Next.
6. Review the Autodesk software license agreement for your country or region *Click I Accept, and then click Next.*
7. On the Personalize the Products page, enter your user information and click Next
8. On the Review - Configure - Install page, click Install to begin installing. By clicking Install
9. Installs AutoCAD to the default install path of *C:\Program Files\AutoCAD*

Basically all Objects are drawn in the XY plane in AutoCAD. A designer can change the current viewpoint to a *plan view* of the current or previous UCS. A *plan view* is a view aimed towards the origin 0,0,0 from a point on the positive Z axis.

AutoCAD Tools and Commands

Various components of the basic AutoCAD screen are the menu bar, drawing area, several toolbars, command window, model and layout tabs and status bar. The title bar has an AutoCAD symbol, and the current drawing name is displayed on top of the screen. The various **AutoCAD Command/tools** present to assist in drawing a certain part/component, the most basic types used are:

Command	Comment/Usage
XLINE	For infinite line
OFFSET	Offset an object by distance (parallel)
INTERSECT	For intersecting lines
ARC	For arc with points
CIRCLE	Draw a Circle
FILLET	Draw an arc between two intersecting lines
LINE	To draw a line (perpendicular lines)
EXTRUDE	To create 3D solid object
TRIM	To trim an object
EXTEND	Boundary object selection

AutoCAD Command/Tools

1. Line: You can invoke the LINE command by choosing the LINE tool from the Draw panel, or you can also invoke the LINE tool by entering LINE or L at the Command Prompt. You will have to specify the starting point of the line by clicking the mouse then you will be prompted to specify the second point. You can terminate the LINE command by pressing ENTER, ESC or SPACEBAR.

2. Circle: A circle is drawn by using the CIRCLE command. You can draw a circle by using six different tools, i.e., by specifying center and radius, by specifying center and diameter, by specifying two diametrical ends, by specifying three points on a circle, tangent to two objects, tangent to three objects.

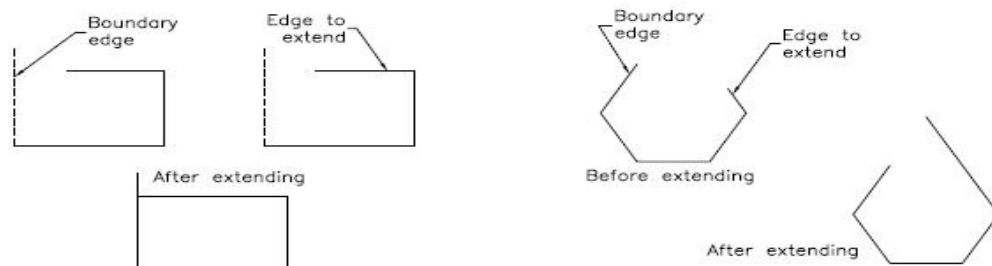
3. Rectangle: You can draw rectangles by specifying two opposite corners of the rectangle, specifying the area and the size of one of the sides, or specifying the rectangle's dimensions.

4. Polyline: Polyline means many lines. To draw a polyline, you need to invoke the PLINE command. After invoking the PLINE command and specifying the start point, the following prompt is displayed:

- ✓ Specify start point: specify the starting point or enter its coordinates
- ✓ Current line width is: nn.nnnn(00.0000)
- ✓ Specify next point or [Arc/Halfwidth/Length/Undo/Width]: specify the endpoint of the first polyline segment.
- ✓ Specify next point or [Arc/Close/Halfwidth/Length/Undo/Width]: specify the endpoint of the second polyline segment or press ENTER to exit the command.

5. Trim: When creating a design, you may need to remove the unwanted and extending edge. In such cases, you can use the Trim tool. On invoking the Trim tool, you will be prompted to select the cutting edges. These edges can be lines, polylines, circles, arcs, ellipses, rays, splines, text, blocks, xlines or even viewports. After the cutting edge/edges are selected, you must select each object to be trimmed.

6. Extend: The Extend tool may be considered as the opposite of the Trim tool. You can extend lines, polylines, rays, and arcs to meet the other objects using the Extend tool. You can use this option whenever you want to extend the objects that do not actually intersect the boundary edge but would intersect its edge if the boundary edges were extended.



7. Copy: This tool is used to make the copies of the selected objects and place them at the specified location. On invoking this tool, you need to select the objects and then specify the base point. Next, you need to specify the second point where the copied objects have to be placed. You can continue specifying the second point for creating multiple copies of the selected entities.

8. Mirror: This tool is used to create a mirror copy of the selected objects. The objects can be mirrored at any angle. This tool is helpful in drawing symmetrical figures. On invoking this tool, you will be prompted to select objects. On selecting objects to be mirrored, you will be prompted to enter the first point of the mirror line and the second point of the mirror line. A mirror line is an imaginary line about which the objects are mirrored.

9. Rotate: On invoking this tool, you will be prompted to select the objects and the base point about which the selected objects will be rotated. By default, a positive angle results in counter clockwise rotation, whereas a negative angle results in a clockwise rotation. The Rotate tool can also be invoked from the shortcut menu by selecting an object and right-clicking in the drawing area, and choosing Rotate from the shortcut menu.

10. Erase: Sometimes, you need to erase the unwanted objects from the objects drawn. To erase an object, choose Erase tool from the Modify panel. To invoke the Modify toolbar, choose View>Windows>Toolbars>AutoCAD>Modify from the ribbon. A small box, known as a pick box, replaces the screen cursor on invoking the Erase tool. To erase the object, select it by using the pick box; the selected object will be displayed in dashed lines, and the Select objects prompt will be displayed again. You can either continue selecting the objects or press ENTER to terminate the object selection process and erase the selected objects.

11. Offset: Offset is one of the 2d commands through which we can make a number of copies of any 2d shape at some distance in its parallel direction for saving our time in making of drawing of that 2d shape again and again. You can use the Offset tool to draw parallel lines, polylines, concentric circles, arcs, curves, etc., While offsetting an object, you need to specify the offset distance and the side to offset.

12. Move: The Move Tool is used to move one or more objects from their current location to a new location without changing their size or orientation.

13. Array: You may need to create an object multiple times in a rectangular or circular arrangement in some cases. This type of arrangement can be obtained by creating an array of objects. In Rectangular Array, you need to mention the number of rows and columns along with the Row offset distance and Column offset distance. Whereas in Polar Array, you need to specify the Centre point around which you need the number of objects.

14. Scale: Sometimes you need to change the size of objects in a drawing. For this purpose, the Scale tool comes in handy.

15. Fillet: The edges in a model are generally filleted to reduce the area of stress concentration. The fillet tool helps form round corners between any two entities that form a sharp vertex.

16. Explode: This tool is useful when you have inserted an entire drawing, and you need to alter a small detail. After you invoke the Explode tool, you are prompted to select the objects you want to explode. After selecting the objects, press ENTER or right-click to explode the selected objects and then end the command.

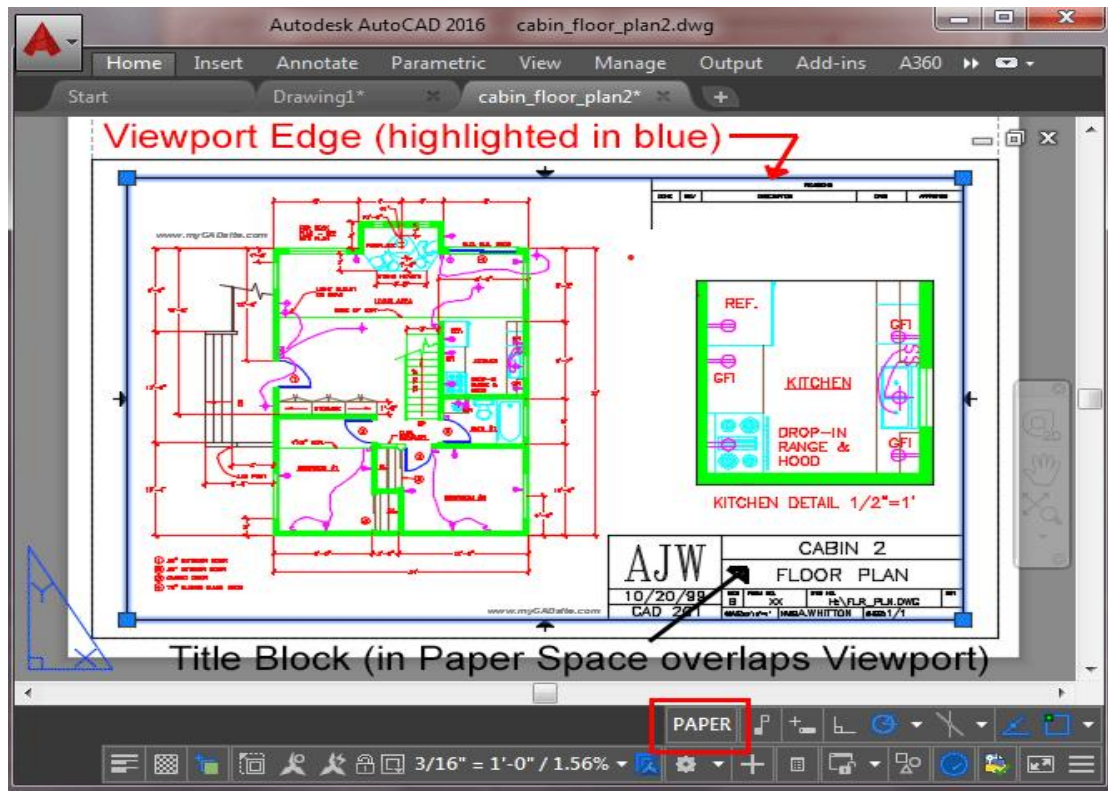
Planning Layout in AutoCAD

Layout in AutoCAD is a 2 dimensional working space for creating drawing sheet. Its purpose is to represent the drawing sheet. Layout area (called “*paper space*”) displays scaled views of model space within the boundaries (called “*viewports*”). In addition, it can contain tables, schedules, notes, annotations for the drawing. A layout typically includes;

- A drawing border and title block.
- One or more layout viewports that display views of model space.
- General notes, labels, and possibly dimensions.
- Tables and schedules.

Main advantages of it is that a drawing can be made in model as full scale 1:1, Layouts have the provisions to display it in any desired scales.

- A layout page allows you to set up a plot or a printout for your drawings.
- A **viewport** is an actual drawing object. As such, you can move, copy, grip-edit, print, and delete it, and change its properties.
- A single **viewport** in a single layout is often appropriate for most drawings. You can add more layouts to the drawing, and each layout can have many viewports.



Plotting - The term **plotting** and printing can be used interchangeably for CAD output. Historically, printers would generate text only, and plotters would generate vector graphics. As printers became more powerful and could generate high-quality raster images of vector data, the distinction largely disappeared.

To plot a drawing in AutoCAD

1. Click Output tab ➤ Plot panel ➤ Plot. Find
2. Select a plotter.
3. Select paper size, plot area, plot scale, orientation and other options.
4. For additional options, click the More Options button.

Title Block -used to name the drawing part, time and author. Essentially, it provides important information for when the document is printed/plotted. inserting a Title Block in your AutoCAD drawing is essential for plotting (printing) your drawings so that it recipients will have the necessary information about them.

Name of company		28		25		15	
		NAME		DATE			
DESIGNED DRAWN CHECKED STANDARD APPROVED DRAUGHTING NO.	15						
SCALE	TITLE	150		SHEET 1 OF 1			

Block

A **block** is a collection of objects (lines, arcs, circles, text, etc.) that form a more complex entity that normally represents an object in the real world, e.g. a door, a chair, a window, a computer, cushion, light bulbs etc. The linework and text making up a block is grouped together in such a way that you can select the entire object (block) at once in order to perform actions such as placing, copying, and moving the block.



- **Blocks are a single entity.** This means that you can modify (move, copy, rotate) a block by selecting only one object in it.
- **You can build up a library of blocks** consisting of the parts that you require many times in your workday. These blocks can be stored in a separate folder and even on a network so that all drafters have access to them. Think of plumbing parts, valves, elbows, etc as well as electrical symbols or furniture.
- **Using blocks can help keep your file size down.** AutoCAD stores block definitions in its database. When you insert a block, AutoCAD only stores the name of the block, its location (insertion point), scale and rotation. This can be very noticeable in large drawing.
- **If you need to change something, you can redefine/Modify a block.** For example, you draw a chair and turn it into a block. Later, you're told that the size of the chair has changed. Since you used a block you can redefine the block and all of your chairs are updated automatically. If you had drawn (or copied) 100 chairs in your drawing, you would have to manually change each one.
- **Blocks can also contain non-graphical information.** These are text objects called **Attributes**. For example, you have made blocks of different chairs. You can add information to the block such manufacturer, cost, weight, material, color etc. This information stays with the block, but can also be extracted to a database or spreadsheet. This would be useful for things such as a **bill of materials**. **Attributes** can also be visible or invisible in your drawing. Another good use of attributes could be a **title block**. You can even easily add internet hyperlinks to blocks so you can connect a block to a page on a supplier's online catalog.

Function of a Block

- Reduce file size.
- Reduce Repetitive task.
- Maintain Company Standard.
- Simplify Library and process
- Creates Report Automatically

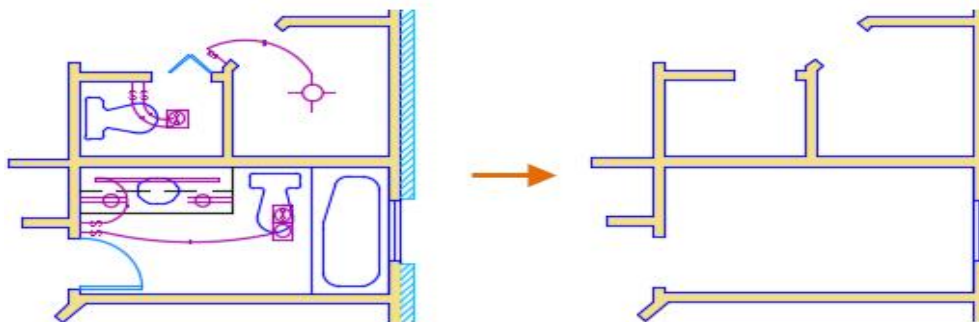
Why you should use Blocks in a Design?

- The main reasons for using the block is its ability to modify all its indicators by changing one block. *For example*, if you use blocks to create **windows floor plan** and after adding windows you decide to change the type of window. In this case, you can simply change the block and all of the indicators used in the drawing will change automatically.

- Blocks also help you keep file size controlled. A drawing made of repetitive blocks will be much smaller than a drawing using copied objects for repetitive objects.
- Blocks also help maintain similarity in the drawings of your project.

Layers in AutoCAD

Layers are the primary method for organizing the objects in a drawing by function or purpose. Layers can reduce the visual complexity of a drawing and improve display performance by hiding information that you don't need to see at the moment. Before you start drawing, create a set of layers that are useful to your work. In a house plan, you might create layers for the *foundation*, *floor plan*, *doors*, *fixtures*, *electrical*, and so on. In this example, the display is limited to the objects on the Floor Plan layer by turning off the other layers. One drawing can have many layers. And depends on the complexity of the drawing,



you may have a few to dozens of layers.

You can use each layer to draw a specific object type. For example, you can use *Wall* layer to draw walls on it. *Furniture* layer to draw furniture on it, and so on. AutoCAD supports tens of thousands layers in a single drawing, which is virtually unlimited.

Why Are Layers Useful ?

- **First**, because you draw objects on separate layers, it is easier for you to understand the drawing. You can quickly know an object type by checking the layer. AutoCAD drawings are just lines and arcs. Using layers can make the drawing easier to understand.
- **Second**, you don't have to set properties for each object. You only need to set the properties for the layers. When you draw an object on a particular layer, it will have the same lineweight, color, linetype, and the other properties. It gives you a better control to plot your drawing.
- **Third**, you can hide/show or lock/unlock a layer. It means you can have greater control of your drawing. You can lock several layers to prevent they are accidentally modified. You can hide objects when the drawing becomes too complicated.
- **Fourth**, using layers adds a better control to select and modify objects. Let's say you want to select all partition walls in your drawing and copy them to the next typical floor. AutoCAD doesn't understand which lines are partition walls. However, we can tell AutoCAD to select lines on the *Partition Wall* layer. Conclusively, AutoCAD layer is the essential drawing management that everyone must use. Regardless the complexity of your drawing. It is not easy to work with a drawing that only has one layer. Everyone hates to receive a drawing like that. Can you imagine how people feel when they don't have a proper control of the drawing because every object are on Layer 0?